COM209 Engineering Mathematics V

Tutorial 5

The joint pmf of two discrete random variables X and Y is given by

$$\frac{\binom{3}{x}\binom{2}{y}\binom{4}{2-x-y}}{\binom{9}{2}}$$

for $x = 0, 1, 2, y = 0, 1, 2, 0 \leq x + y \leq 2$.

(a) Find the marginal probability mass functions $p_X(x)$ and $p_Y(y)$.

(b) Determine the value of $F(1, 1)$.

(c) Determine the values of $F_X(1)$ and $F_Y(1)$.

Let

$$F(x,y) = \begin{cases} \dfrac{y + e^{-x(y+1)}}{y+1} - e^{-x} & x > 0, y > 0 \\ 0 & \text{otherwise} \end{cases}$$

(a) Find the joint density $f(x,y)$.

(b) Find the marginal densities of $f(x,y)$.

6.22. The joint density function of X and Y is

$$f(x,y) = \begin{cases} x + y & 0 < x < 1,\ 0 < y < 1 \\ 0 & \text{otherwise} \end{cases}$$

(a) Are X and Y independent?
(b) Find the density function of X.
(c) Find $P\{X + Y < 1\}$.

(a) Let X, Y, and Z be the number of hits at a website on three consecutive days. Assuming they are independent Poisson(λ) random variables, find the probability that on each day the number of hits is at most n.

(b) A webpage server can handle r requests per day. Find the probability that the server gets more than r requests at least once in n days. Assume that the number of requests on day i is $X_i \sim Poi(\lambda)$, and that $X_1, X_2, \ldots, X_n$ are independent.